Figure 3.1: The Gram-Schmidt Procedure. Given two linearly independent vectors, $\vec{V}_1$ and $\vec{V}_2$, that are not necessarily orthogonal, we can construct two orthonormal vectors, $\hat{\mathbf{v}}_1$ and $\hat{\mathbf{v}}_2$. $\vec{V}_{2\perp}$ is an intermediate result used in the construction process. We can extend this procedure to larger sets of linearly independent vectors.

The first step is to divide $\vec{V}_1$ by its own length, $|\vec{V}_1|$, which gives us a unit vector parallel to $\vec{V}_1$. We'll call that unit vector $\hat{\mathbf{v}}_1$, and $\hat{\mathbf{v}}_1$ becomes the first vector in our orthonormal set. Next, we project $\vec{V}_2$ onto the direction of $\hat{\mathbf{v}}_1$ by forming the expression $<\vec{V}_2|\hat{\mathbf{v}}_1>\hat{\mathbf{v}}_1$. Now, we subtract $<\vec{V}_2|\hat{\mathbf{v}}_1>\hat{\mathbf{v}}_1$ from $\vec{V}_2$. We'll call the result of this subtraction $\vec{V}_{2\perp}$. You can see in Fig. 3.1 that $\vec{V}_{2\perp}$ is orthogonal to $\hat{\mathbf{v}}_1$. Lastly, we divide $\vec{V}_{2\perp}$ by its own length to form the second member of our orthonormal set, $\hat{\mathbf{v}}_2$. It should be clear that we can extend this procedure to larger sets of linearly independent vectors in more dimensions. For instance, if we had a third linearly independent vector, say $\vec{V}_3$, pointing out of the page, we would subtract its projections onto each of the unit vectors $\hat{\mathbf{v}}_1$ and $\hat{\mathbf{v}}_2$, and then divide the result by its own length.[1]

3.2 THE PRINCIPLES

We are now fully prepared to state the principles of quantum mechanics, so without further ado, let's do it.

The principles all involve the idea of an observable, and they presuppose the existence of an underlying complex vector space whose vectors represent system states. In this lecture, we present the four principles that do not involve the evolution of state-vectors with time. In Lecture 4, we will add a fifth principle that addresses the time development of system states.

An observable could also be called a measurable. It's a thing that you can measure with a suitable apparatus. Earlier, we spoke about measuring the components of a spin, σ_x, σ_y, and σ_z. These are examples of observables. We'll come back to them, but first let's look at the principles:

- Principle 1: The observable or measurable quantities of quantum mechanics are represented by linear operators **L**.